

Types of data

What type of data is each of:

- A firm conducts a survey of 500 students about their current health habits

- A program tracks the same 10 students every month over their entire college career, measuring their health habits

- For 5 years in a row, each incoming freshman class is surveyed on their health habits

- A student records their health habits every day for a year

First differences

The table below has annual sales for a company over five years. Calculate the first difference of each available period.

Table 1: Annual sales (\$1000s)					
Year	2021	2022	2023	2024	2025
Sales (Y_t)	120	145	130	160	175

Based on the first differences, was sales growth constant over these five years?

You have two periods of data for 100 firms, with their sales and marketing budget. Suppose, after first differencing, we regress the log sales on log marketing budget. We estimate $\beta_1 = 0.14$. Interpret this coefficient.

Fixed effects

Why can't we estimate the effect of time-invariant variables (such as gender or race) using fixed effects? Try it: write a regression equation with a time-invariant variable (only has i in the subscript) and see what happens when you subtract out the group's mean of each variable.

Table 2: Wages and training

Worker	Year	Certifications completed (X_{it})	Wage (Y_{it})
1	1	2	20
1	2	4	24
1	3	6	26
2	1	3	30
2	2	5	34
2	3	7	36

Apply de-meaning to both variables and find the fixed effects regression coefficient.

Recall: $Y_{it} = \alpha_i + \beta_1 X_{it} + u_{it}$, $\beta_1 = \frac{\mathbb{E}[(Y_{it} - \bar{Y}_i)(X_{it} - \bar{X}_i)]}{\mathbb{E}[(X_{it} - \bar{X}_i)^2]}$

The pooled OLS estimate from the regression of wages on certifications is 2.23. Is your answer different with effects? If so, why might this be? (hint: what question does pooled OLS answer, and what question does FE answer)